

CAITS — Product Functional Guide



CAITS (Centralized Asset and Inventory Tracking System) helps teams manage inventory and serialized assets across organizations, operating units, and store locations — from receipt and inspection through issue, transfer, return, and gatepass.

 Frontend	React + Vite (<code>http://localhost:5173</code>)
 Backend	Spring Boot + JWT (<code>http://localhost:8085/api/v1</code>)
 Database	PostgreSQL schema <code>caits_local</code>

This guide focuses on **flows and functionality**. For schema/API depth, see `CAITS_FULL_SYSTEM_DOCUMENTATION.md` .

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1. Who uses it



Icon	Role area	Typical use
	Store / inventory	Opening stock, GRN, issue, transfer, return, stock ledgers
	Gate / logistics	Gatepass inward & outward, returnable materials
	Quality	Inspection approval (quarantine → home store / rejected)
	Admin	Masters, roles, users, location access, exceptions

Access is controlled by **Role & Menu Mapping** (view / create / edit / delete / approve / print / export), plus optional **User Access Exceptions** and **OU / location scope** on the user.

flowchart LR

```

U[  User login] --> R[  Role]
R --> M[  Menu ACL]
U --> S[  OU / Location scope]
M --> A[Allowed screens & actions]
S --> A

```

2. Core concepts

2.1 Assets vs consumables

 Assets vs consumables comparison

Type	Tracking	Typical behaviour
 Asset	Often serialized (1 unit = 1 serial)	Serial on receive/issue/return/outward; custody on BLS
 Consumable	Qty / optional batch	Stock quantity increases/decreases at locations

2.2 Stock vs BLS

flowchart TB

```

subgraph Stock[" Stock register"]
    S1["Item + Location (+ batch)"]
    S2["Quantity on hand"]
end
subgraph BLS[" BLS register"]
    B1["Serial / lot / unit"]
    B2["Location · Issued-to · IP/MAC"]
end

```

```
Stock -. qty view .-> Ops[Transactions]
BLS -. unit view .-> Ops
```

Register	What it answers
Stock (inv_stock_mst)	<i>How much</i> is at a location?
BLS (inv_bls_mst)	<i>Which unit / serial</i> is where, and who holds it?

2.3 Organization & system locations

 Organization hierarchy and system locations

System location	Purpose
 MAIN_STORE	Default system store
 QUARANTINE	Inspection-needed receipts wait here
 DAMAGED	Damaged material
 SCRAP	Scrap
 REJECTED	Rejected at GRN or after failed inspection

Operational **Locations** belong to Operating Units and are used for day-to-day store work.

2.4 Inspection needed

 Inspection needed workflow

On **Item Master**, set **Inspection Needed**. On GRN / Gatepass Inward submit:

1. Accepted qty → **Quarantine**
2. **Pending Inspection Approval** for the assigned inspector
3. **Approve** → item **home store** · **Reject** → **Rejected**

2.5 Document lifecycle

 Document lifecycle: draft, submit, final

Action	Result
 Save Draft	Editable draft / pending
 Submit	Posts stock (where applicable) and finalizes

Document numbers: PREFIX-YEAR-NNNN — e.g. GRN-2026-0001 , GPO-2026-0003 .

3. Master setup (recommended order)

Set masters in this order so transactions have the data they need:

```
flowchart LR
    A[1 Units] --> B[2 Categories]
    B --> C[3 General Master]
    C --> D[4 Org → OU → Location]
    D --> E[5 Items]
    E --> F[6 Vendors]
    F --> G[7 Roles]
    G --> H[8 Employees]
    H --> I[9 Users]
    I --> J[10 Exceptions]
```

Step	Master	Why
1	Unit Master	UOMs
2	Inventory Category / Sub-Category	Item classification
3	General Type / General Master	Lookups (returnable, purposes...)
4	Organization → OU → Location	Structure + system stores
5	Item Master	Type, serial/batch, home store, inspection
6	Vendor / Party	Suppliers & parties
7	Role & Menu Mapping	Screen permissions
8	Employee / Department	People & teams
9	User Access Mapping	Login, role, scope
10	User Access Exception	Optional overrides

4. End-to-end business flows

4.1 High-level stock journey

End-to-end stock journey diagram

```
flowchart TD
    OS[Opening Stock] --> ONHAND[Stock / BLS on hand]
```

```

GRN[GRN] --> ONHAND
GPI[Gatepass Inward] --> ONHAND
ONHAND -->|Inspection needed| Q[Quarantine]
Q --> IAPR[Inspection Approval]
IAPR -->|Approve| HOME[Home store]
IAPR -->|Reject| REJ[Rejected]
ONHAND --> SR[Requisition]
SR --> ISS[Store Issue]
ISS --> RTN[Material Return]
ONHAND --> TRF[Material Transfer]
TRF -->|OU| PFO[Pending for Outward]
PFO --> GPO[Gatepass Outward]
GPO -->|Returnable| GPI2[Gatepass Inward]

```

4.2 📦 Opening Stock (OPN)

Purpose: Seed opening balances.

Step	Action
1	Create entry: organization, store, date, item lines
2	Assets → serials · Consumables → qty (+ optional batch)
3	Submit → stock increases ; BLS created/resolved

⚠️ Duplicate serials already in the system are rejected (first-time inbound).

4.3 📄 Goods Receipt Note — GRN

Purpose: Receive material from a **supplier / vendor**.

Step	Action
1	Header: supplier, invoice/PO refs, store, inspector if needed
2	Lines: received / accepted / rejected qty; serials for assets
3	Submit

Where accepted / rejected qty goes:

flowchart LR

SUB[Submit GRN] --> REJ[ Rejected qty → Rejected store]

SUB --> ACC{Accepted?}

ACC -->|Inspection needed| Q[ Quarantine + IAPR]

ACC -->|Normal| ST[ GRN / line store]

Qty	Destination
Rejected	Rejected system store
Accepted + Inspection Needed	Quarantine (+ pending Inspection Approval)
Accepted otherwise	GRN / line store

Requires: Inspected By when any accepted line needs inspection; item must have a **home store**.

4.4 Inspection Approval (IAPR)

Purpose: Clear quarantine after GRN or Gatepass Inward.

Step	Action
	System creates Pending approval linked to source document
	Assigned inspector opens Inspection Approvals
	Approve → Quarantine → home store
	Reject → Quarantine → Rejected store

 Only the assigned inspector (or ADMIN) may approve/reject.

4.5 Store Requisition (SR)

Purpose: Request material — **does not move stock**.

Step	Action
	Return-from type: Employee or Department
	Select items/locations with available stock
	Submit → status Requested

Downstream: Store Issue picks **Requested** requisitions → after issue, requisition becomes **Issued**.

4.6 📦 Store Issue (ISS)

 Request → Issue → Return cycle

Purpose: Issue material from a store against a requisition.

Step	Action
1	New Issue → pick a Requested requisition
2	Confirm From Location, Issued To, To Location
3	Lines: issue qty; assets need free serials at From Location
4	Submit → Issued

Item type	On submit
 Consumables	Quantity leaves the issuing store
 Assets (same store)	On-hand qty often unchanged; BLS issued-to employee
 Different destination	Stock moves From → To

Sidebar highlights **Store Issue** for all `/transactions/issues/*` routes.

4.7 🔄 Material Transfer (TRF)

Purpose: Move stock between locations.

Type	Behaviour
 INTERNAL	Submit → stock moves → Transferred
 OU	Submit → stock moves → Pending for Outward until Gatepass Outward → Transferred

```
sequenceDiagram
    participant User
    participant Transfer as Material Transfer
    participant Stock
    participant Gate as Gatepass Outward
    User->>Transfer: Submit OU transfer
    Transfer->>Stock: Move qty / BLS
    Transfer-->>User: Pending for Outward
```

User->>Gate: Complete outward
Gate->>Transfer: Status → Transferred
Note over Gate: Gate docs only – no second stock post

Rules:

- Cannot transfer inspection-needed items **into Damaged/Scrap** (use Quarantine / inspection).
- Linked Gatepass Outward is **gate documentation only** (does not post stock again).

4.8 Material Return (RTN)

Purpose: Return allotted material to store.

Step	Action
1	Return From: Employee or Department
2	Pick allotted item (and serial for assets)
3	Return-to store = Item Master home store
4	Submit → Returned ; stock back; BLS custody cleared

4.9 Gatepass (GP)

One menu covers **Outward** and **Inward**.

 Gatepass and OU transfer / returnable cycle

Outward

Mode	Purpose
New / standalone	From a system location to a vendor/party; Returnable Y/N
Against transfer	Completes OU transfer; party/store from transfer

Rule	Detail
 Mandatory	Vendor / Party / Customer (free text)
 Prefill	Party can come from source GRN / Opening Stock vendor
 Block	Inspection-needed items from Quarantine / Damaged / Scrap until inspected

Mode	Purpose
Against Returnable Outward	Bring returnable material back
New Inward	Fresh gate receipt from Item Master

Rule	Detail
 Mandatory	Vendor / Party / Customer · Serial for assets
 Inspector list	Employees who have a role assigned

Receive location:

Situation	Location
New inward, no inspection	Item home store
Inspection needed (new or returnable)	Quarantine (+ Inspection Approval)
Returnable, no inspection	Outward system store

5. Reports & Dashboard

Icon	Screen	What you get
	Dashboard	KPIs, pending work, low stock, recent activity
	Stock Ledger	Opening / receipts / issues / closing; expand units
	Log Report	Transaction activity (filters include serial)
	Stock Owner Report	Where stock sits and who holds custody
	Asset Movement Register	How assets moved across locations/custodians
	Item Ledger	Running balance by document for an item

flowchart LR

D[Dashboard] --> P[Pending work]

D --> L[Low stock]

R1[Stock Ledger] --> Bal[Balances]

R2[Stock Owner] --> Cust[Custody]

R3[Asset Movement] --> Trail[Movement trail]

6. Common business rules (cheat sheet)

Icon	Topic	Rule
	Serials	Required for serialized/asset lines on receive, issue, return, gatepass
	Duplicate serial	Blocked on GRN / Opening Stock; Return & GPI reuse BLS
	Inspection	Quarantine → Inspection Approval → Home or Rejected
	Gatepass party	Mandatory on inward and outward
	OU transfer	Needs Gatepass Outward to finish
	Returnable outward	Paired later with Gatepass Inward
	Location access	Users only see/post to allowed OUs/locations (ADMIN exempt)
	Drafts	Editable until submitted

7. Suggested demo walkthrough

Follow this path for a short end-to-end demo:

```
flowchart LR
  M[1 Masters] --> R[2 Receive]
  R --> Q[3 Inspect]
  Q --> I[4 Req + Issue]
  I --> T[5 Return]
  T --> G[6 Transfer + GP]
  G --> V[7 Verify reports]
```

Step	What to do
1	Masters — org, OU, location, units, one asset + one consumable, vendor, employee with role + login
2	Opening Stock or GRN — receive asset with serial and consumable qty
3	If inspection-needed — Inspection Approval → home store
4	Requisition → Store Issue (pick serial for asset)
5	Material Return — return the allotted asset
6	Transfer (OU) → Gatepass Outward (returnable) → Gatepass Inward
7	Open Stock Ledger / Stock Owner / Dashboard to verify

8. Related docs in this repo

File	Content
<code>CAITS_PRODUCT_FUNCTIONAL_GUIDE.md</code>	This guide (flows, diagrams, images)
<code>images/</code>	Cover, role art, and SVG flow diagrams
<code>CAITS_FULL_SYSTEM_DOCUMENTATION.md</code>	Deeper technical / schema write-up
<code>DEMO_UI_WORKFLOW_RUN.md</code>	Demo UI workflow notes
<code>../db_approach/CAIMS_Relational_Schema.md</code>	Relational schema notes
<code>../README.md</code>	Project entry / run instructions

Viewing tips

Viewer	What works best
GitHub / GitLab	PNGs, SVGs, and Mermaid diagrams
VS Code / Cursor	Open preview (<code>Markdown: Open Preview</code>) for Mermaid + images
PDF export	Use a Markdown→PDF tool that supports Mermaid (or rely on the SVG images)

Aligned with current CAITS behaviour (masters, transactions, quarantine/inspection, gatepass, reports). Update this file and `docs/images/` when major workflows change.